Linux Networking Fundamentals

DEVOPS-003 – Day 3 Deliverable: Networking Theory

Day3_Networking_Theory

Goal: *Understand how Linux communicates with other systems.*

1. What is a Computer Network?

A computer network is simply two or more devices – computers, servers, printers, phones – connected together so they can share data and resources. The easiest way to picture it is a group of people connected by phone lines: each one can talk to the others, pass along files, or share something like a printer, without needing to be in the same room.

Networks come in different sizes depending on how far they reach:

- **LAN (Local Area Network)** – a small network, like the Wi-Fi in a home or office.
- **WAN (Wide Area Network)** – a network that spans a large area; the internet is the biggest example.
- **MAN (Metropolitan Area Network)** – covers something in between, like a city or a large campus.

This matters in DevOps because servers, containers, and services are almost never isolated – they are constantly talking to each other over a network, and a large part of troubleshooting comes down to one question: can this machine actually reach that machine?

2. What is an IP Address?

An IP address (Internet Protocol address) works like a home address for a device on a network. Every device that wants to send or receive data needs one, so that other devices know exactly where that data should go.

There are two versions in use today:

- **IPv4** – written as four numbers separated by dots, such as 192.168.1.10, with each number ranging from 0 to 255. This is still the most common format.
- **IPv6** – a newer, much longer format (for example 2001:0db8:85a3::8a2e:0370:7334), introduced mainly because the world is running out of available IPv4 addresses.

Without an IP address, a computer has no way to identify itself on a network or know where to send a reply – it is essentially the device's identity for communication.

3. Difference between Public IP and Private IP

This distinction confuses a lot of people at first, but it is fairly simple once it clicks:

- **Private IP** – the address a device uses inside its own local network, such as a home or office Wi-Fi. It is only visible to other devices on that same network. Common private ranges include 192.168.x.x, 10.x.x.x, and 172.16.x.x through 172.31.x.x. Two completely different homes can each have a device using 192.168.1.5 at the same time with no conflict, because private IPs are never exposed to the outside world directly.
- **Public IP** – the address a network uses to communicate with the wider internet. It is assigned by an Internet Service Provider (ISP) and is unique across the entire internet. The router is usually the device holding this public IP, and it translates traffic between the private devices inside the network and the public internet through a process called NAT (Network Address Translation).

A simple way to remember it: the private IP is like an apartment number, and the public IP is like the street address of the whole building. Data reaches the building using the street address, then gets routed internally to the right apartment.

4. What is a MAC Address?

A MAC address (Media Access Control address) is a unique hardware identifier assigned to a device's network interface at the factory. Every network card, whether it is in a laptop, phone, or server, has one, and it looks something like 00:1A:2B:3C:4D:5E – six pairs of hexadecimal digits.

The key differences between a MAC address and an IP address:

- A MAC address is tied to the physical hardware and normally never changes.
- An IP address is tied to the network and can change depending on where the device connects.
- MAC addresses operate at Layer 2 (Data Link layer), while IP addresses operate at Layer 3 (Network layer).

A useful way to think about it: the MAC address is the device's permanent serial number for networking, while the IP address is more like its current location, which can change from network to network.

5. What is DNS?

DNS stands for **Domain Name System**, and it is often described as the phonebook of the internet.

Computers communicate using IP addresses, but people are far better at remembering names than long strings of numbers. Nobody wants to type 142.250.183.14 to visit a website when they could just type google.com instead. DNS is the system that translates those human-friendly domain names into the machine-friendly IP addresses computers actually need.

In practice, this happens in three quick steps whenever a website is opened in a browser:

- The computer asks a DNS server, “what is the IP address for this domain?”
- The DNS server looks it up and sends back the matching IP address.
- The browser then connects directly to that IP address to load the site.

All of this typically happens in a fraction of a second, well before a person notices any delay.

6. What is a Gateway?

A gateway, usually called the **default gateway**, is the device – typically a router – that sits between a local network and the outside world, such as the internet.

Here is the practical picture: when a device wants to send data somewhere outside its own network, it has no direct way of reaching a far-off server on its own. So it hands the data to the gateway first, and the gateway takes care of forwarding it onward, out to the internet, and routing any response back.

A simple analogy is a building's front door – everything that goes in or out of the network has to pass through it.

7. What is a Port Number?

If an IP address is like a building's street address, a **port number** is like a specific apartment or office number inside that building. It tells a computer exactly which service or application incoming data is meant for.

A single machine can run many services at once – a web server, a mail server, an SSH service, and more – and port numbers are what let all of them run at the same time without their traffic getting mixed up. Ports range from 0 to 65535, and a handful of low-numbered ports are reserved for well-known services:

- **Port 22** – SSH (secure remote login)
- **Port 80** – HTTP (regular web traffic)
- **Port 443** – HTTPS (secure web traffic)
- **Port 21** – FTP (file transfer)
- **Port 25** – SMTP (sending email)

So when a browser connects to a website, it is really connecting to an IP address on a specific port – almost always port 443 for a secure site.

8. What is SSH?

SSH stands for **Secure Shell**. It is a protocol that allows secure login to, and control of, a remote computer over a network – something that comes up constantly in Linux and DevOps work, since servers are usually managed remotely rather than sitting in front of the person using them.

Before SSH became standard, tools like Telnet sent everything, including passwords, as plain, unencrypted text, so anyone watching the network traffic could read it. SSH solved this by encrypting the entire connection, meaning that even if the traffic is intercepted, it cannot be read.

A typical SSH command looks like this:

```
ssh username@server_ip_address
```

This is the standard way DevOps engineers connect to remote Linux servers to run commands, move files, deploy applications, and manage infrastructure – all without the connection being exposed.

9. What is HTTP vs HTTPS?

HTTP and HTTPS are both protocols used to move data between a browser and a website's server. The difference between them comes down to security.

- **HTTP (HyperText Transfer Protocol)** – sends data as plain text. If someone intercepts the traffic, they can read everything in it, including sensitive details like passwords or card numbers.
- **HTTPS (HyperText Transfer Protocol Secure)** – works the same way as HTTP, but the data is encrypted using SSL/TLS. Even if the traffic is intercepted, all an attacker sees is scrambled, unreadable data.

The difference is usually visible right in the browser's address bar: HTTPS sites show a padlock icon, while HTTP-only sites are commonly flagged as “Not Secure.” For any site that handles logins, personal details, or payments, HTTPS is no longer optional – it is the expected standard.